

Experiment 2: Exploring Packet Sniffer and Packet Analyzer.

Question 1.

What are the different protocols you observe at the following layers of the protocols stack?

- A. Application layer
- B. Transport layer
- C. Network layer

→ Protocols observed at the application layer are:-

- (i) HTTP (Hypertext transfer protocol)
- (ii) DNS (Domain Name System)
- (iii) SSDP (Simple Service Discovery protocol)

→ At Transport layer are:-

- (i) TCP (Transmission Control protocol)
- (ii) UDP (User Datagram protocol)
- (iii) TLS (Transport layer Security)

→ At Network layer protocols are:-

- (i) IP v4 (Internet protocol version 4)
- (ii) IP v6 (Internet protocol version 6)
- (iii) ARP (Address Resolution Protocol)

Question 2.

What is the total amount of data being received for the following two cases?

- A. when you access <https://neverssl.com/>
- B. when you access <https://www.cornell.edu>

- While accessing 1st, I received data of 848 bytes. (4 packets)
And while accessing 2nd, I received data of 14 MB. (10,890 packets).

Question 3.

A. How many HTTP GET requests do you observe? List down the GET requests.

→ 3 HTTP GET requests.

1. /online/

2. /online

3. /favicon.ico

B. What version of HTTP the server is running?

→ HTTP / 1.1.

C. (i) The total number of TCP segments being received:

→ 7.

(ii) The total amount of data being received in corresponding HTTP response message.

→ Seq = 12643, Ack = 9456, Win = 61440

Q.

(i) SYN → The process two computers used to establish reliable connection before transferring.

(ii) SYN-ACK → Establish a reliable error-free connection between client & server.

(iii) ACK → fundamental networking process.